STAT 361 Assignment 2 Question 2

Edward Tanurkov

2026-10-13

```
dat=iris
```

a) Outlier Function Definition

```
outliers <- function(data) {
  q1 <- quantile(data, 0.25, names = FALSE)
  q3 <- quantile(data, 0.75, names = FALSE)
  iqr = q3 - q1

  lower_bound <- q1 - 1.2 * iqr
  upper_bound <- q3 + 1.2 * iqr

  outlier_vals <- data[data < lower_bound | data > upper_bound]
  num_outliers <- length(outlier_vals)

  if (num_outliers == 0) {
    outlier_vals <- NULL
  }

  return(list(num_outliers = num_outliers, outlier_vals = outlier_vals))
}
```

b) Outlier Function Application

```
# Loop through each unique species from the iris dataset
for (sp in levels(iris$Species)) {
  subset_data <- subset(iris, Species == sp)
  vectorized_data <- subset_data$Petal.Width
  found_outliers <- outliers(vectorized_data)

  cat(paste0("Number of Outliers for ", sp, ": ", found_outliers$num_outliers, "\n"))
  cat("Set of Outliers: ")
  print(found_outliers$outlier_vals)
  cat("\n")
}
```

```
## Number of Outliers for setosa: 2
```

```
## Set of Outliers: [1] 0.5 0.6
##
## Number of Outliers for versicolor: 0
## Set of Outliers: NULL
##
## Number of Outliers for virginica: 0
## Set of Outliers: NULL
```

There are two outliers for Petal Width in the setosa species. This is also seen in the respective boxplot in Q1 a). The other two species do not have outliers present in their data, also as seen in the Q1 boxplot.